

Akarsh Jain

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PROFILE

Backend-focused Computer Science undergraduate at SVNIT Surat who ships production systems – FastAPI/PostgreSQL, Docker, AWS EC2 – backed by automated tests and real CI/CD. Seeking a Software Engineering / Backend Development internship.

EDUCATION

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat

Bachelor of Technology (B.Tech), Computer Science and Engineering

2024 – 2028

CGPA: 7.55 / 10

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, TypeScript, SQL

Backend & APIs: FastAPI, REST API Design, JWT/OAuth2, LangGraph, Pydantic, RBAC

Testing & Tools: Pytest, Unit & Integration Testing, Git, GitHub Actions, Docker, Postman

Deployment & Infra: AWS EC2, Render, Caddy, Docker Compose, CORS, Env Configuration

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

PROJECTS

Campus Resource Sharing System – Peer-to-Peer Lending for Campuses

GitHub – Live Demo

- FastAPI, SQLAlchemy 2.0, PostgreSQL, Alembic, Redis, Celery, JWT/RBAC, React 18, Tailwind CSS, Docker Compose
- A campus-only marketplace where students, faculty, and clubs list, borrow, and return shared resources; models the full borrow lifecycle (request → approve/reject → return → review) across 4 RBAC roles, with WebSocket real-time notifications and Celery-beat return reminders.
- Shipped 17 REST routers on FastAPI/SQLAlchemy 2.0 with CSRF-hardened, Redis rate-limited auth, covered by 70+ Pytest cases, and containerized the full stack via Docker Compose.

SentinelReview – Agentic Security Code Review for GitHub PRs

GitHub

- Python, FastAPI, LangGraph, PostgreSQL/pgvector, Redis/RQ, React, TypeScript, Bandit, Semgrep
- An AI reviewer bot that scans GitHub PRs for security vulnerabilities via a 7-agent LangGraph pipeline (Bandit/Semgrep + TF-IDF/cosine retrieval), explaining each finding against CWE/OWASP sources and proposing a verified patch; GitHub App JWT (RS256) + OAuth2 auth and HMAC webhooks trigger RQ workers that persist findings end-to-end.
- Reached 0.900 precision / 1.000 recall (0.952 F1) on a hand-labeled benchmark after ingesting 150 live GHSA advisories, and cut analyzer latency 5x via batching, backed by 101 backend + 20 frontend tests.

IronLog – Gym Progress Analytics Platform

GitHub – ironlog.in

- FastAPI, SQLAlchemy, JWT, Vanilla JS PWA, Chart.js, Docker, AWS EC2, Caddy, GitHub Actions
- A multi-user fitness tracker that computes Epley 1RM, Mifflin-St Jeor BMR/TDEE, and a true maintenance-calorie figure back-calculated from each user's own logged intake vs. real weight change, plus regression-based weight-trend and goal-ETA forecasting.
- Deployed on AWS EC2 via Docker Compose behind Caddy (auto SSL) with GitHub Actions CI/CD; installable PWA with a Gemini-powered AI Coach and period-over-period progress comparisons, covered by 35 Pytest tests.

Mini Code Judge

GitHub – Live Demo

- FastAPI, PostgreSQL, Docker, Redis/RQ, JWT, Gemini API
- A competitive-programming judge in the spirit of Codeforces/LeetCode that runs untrusted C/C++/Java/Python submissions in resource-capped, network-isolated Docker sandboxes with an automatic OS-level fallback, returning verdicts (AC/WA/TLE/MLE/RE/CE) via Redis/RQ-queued, horizontally-scalable workers.
- Multi-provider auth (email verification, Google, GitHub OAuth) with account lockout; timed contests with live leaderboards and penalty tracking, plus Gemini-powered AI code review after each submission.

ACHIEVEMENTS

Secured 1st Rank in ACM Summer Challenge, August 2025.

Participated in national-level coding contests including Epiphany 14.0 and CodeCraft 2025.

COMPETITIVE PROGRAMMING

LeetCode: 700+ Problems Solved

Profile

Codeforces: Rating 1850+

Profile

CodeChef: Rating 1670+

Profile

INTERESTS

Competitive Programming, Software Development, Artificial Intelligence, DevOps and Cloud Infrastructure, Fitness